

SANTHOSH V S

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Summary

- Results-driven Senior PHP Developer with 10+ years of experience building scalable web applications and SaaS platforms across Healthcare, E-Commerce, Education, Billing, and ERP domains.
- Architected multi-tenant SaaS platforms using Laravel with tenant-aware database connections via spatie/laravel-multitenancy isolating 500+ client datasets within a single application deployment.
- Engineered Laravel Horizon-managed Redis queue workers processing asynchronous invoice generation, payment webhook handling, and scheduled billing jobs with retry policies and failure alerting.
- Built Laravel API resources with Fractal transformers standardising JSON response structures, field inclusion via sparse fieldsets, and pagination envelopes consumed by React and Vue.js frontend clients.
- Designed MySQL schemas with composite indexes, covering indexes, and EXPLAIN-driven query optimisation reducing complex financial report generation from 8 seconds to 340 milliseconds under production load.
- Implemented Laravel Sanctum API token authentication with ability-scoped tokens enabling fine-grained permission control per API consumer (mobile app, third-party partner, internal service).
- Integrated Stripe Billing with webhook signature validation, idempotency key management, and subscription lifecycle event processing (customer.subscription.updated, invoice.payment_failed) via Laravel queue.
- Built CodeIgniter ERP modules with stored procedures for complex inventory valuation (FIFO, LIFO, weighted average) ensuring accurate cost of goods sold calculation across multi-warehouse operations.
- Implemented Laravel Scout with Meilisearch for full-text product and patient search delivering sub-50ms typeahead results across catalogues of 500,000+ records without MySQL LIKE query overhead.
- Conducted code reviews and mentored 5 junior developers on SOLID principles, Laravel service container patterns, and database normalisation ensuring production-grade code quality across all projects.
- Participated in full SDLC from requirement analysis to production deployment managing releases on Linux cPanel and VPS environments with zero-downtime deployment using Laravel maintenance mode and queue restart.

Skills

Languages PHP (Advanced), JavaScript, HTML5, CSS3, SQL, AJAX	Integrations Stripe (Billing, Webhooks, Connect), Razorpay, PayPal, Twilio (SMS), Mailgun, AWS S3, Meilisearch
Frameworks & Libraries Laravel (Advanced), CodeIgniter, Bootstrap, jQuery, Laravel Horizon, Laravel Scout, Livewire	Tools & Platforms Git, GitHub, Postman, Composer, cPanel, Linux/VPS, Supervisor, PM2
Backend Architecture Multi-Tenant SaaS (spatie/laravel-multitenancy), REST API Design, Service Container, Repository Pattern, Queue Workers (Redis, Horizon), Webhooks, Caching	Domains SaaS Platforms, E-Commerce, Healthcare, Billing & ERP, Education Systems

(Redis, Memcached)

Database MySQL (Schema Design, Query Optimisation, EXPLAIN, Stored Procedures, Composite Indexes, Transactions), Redis

Experience

Senior Web Developer Veuston International IT Services, Kollam, Kerala August 2025 – Present

Senior Web Developer Texila Educare Healthcare & Technology Pvt Ltd, Coimbatore August 2024 – July 2025

Senior Web Developer LezDo Tech Pvt Ltd, Kanyakumari August 2023 – June 2024

Senior Web Developer CyberFinite Software Solutions, Chennai November 2022 – July 2023

Web Developer Digitalthink Innovations Networks, Trivandrum January 2015 – October 2022

Education and Training

Master of Computer Applications (MCA)

CMS College, Bharathiar University

Bachelor of Science (Computer Science)

VTM College, MS University

Project

PROJECT 1: BillingFlow – Chargebee-Style SaaS Billing Platform | Domain: FinTech / SaaS Billing

Client: Paddle (UK-based SaaS subscription billing platform) | **Domain:** FinTech / Subscription Management

Tech Stack: PHP, Laravel, MySQL, Redis, Laravel Horizon, Stripe Billing, REST API, Bootstrap, Git, Linux

Description: Multi-tenant SaaS billing and subscription management platform enabling 3,000+ software businesses to manage subscription plans, proration billing, dunning workflows, tax calculations, and invoice reconciliation.

Roles & Responsibilities:

- Architected multi-tenant SaaS database isolation using spatie/laravel-multitenancy with tenant-aware Eloquent scopes and database connection switching per HTTP request for 3,000+ client datasets.
- Implemented Laravel Horizon Redis queue workers processing subscription lifecycle events: plan upgrades, proration calculations, dunning retries (3-day, 7-day, 14-day), and invoice PDF generation asynchronously.
- Integrated Stripe Billing API with webhook signature validation (Stripe-Signature header HMAC-SHA256), idempotency keys on payment requests, and event handlers for customer.subscription.updated and invoice.payment_failed.
- Built subscription management module covering plan configuration, billing cycle scheduling, mid-cycle upgrades with credit calculation, pause/resume, and cancellation with pro-rated refund computation.
- Designed MySQL billing schema with covering indexes on (tenant_id, subscription_status, next_billing_date) reducing monthly billing batch query from 12 seconds to 280 milliseconds for 50,000+ active subscriptions.
- Implemented Laravel Scout with Meilisearch indexing client, subscription, and invoice records enabling sub-50ms full-text typeahead search across 500,000+ billing records without MySQL LIKE query performance degradation.

- Built automated invoicing workflow: Laravel scheduled command generates invoices nightly, dispatches PDF generation job to Horizon queue, stores PDF on S3, and triggers Mailgun email delivery with signed download link.
- Developed role-based access control using Laravel Gates and Policies distinguishing Super Admin, Billing Manager, Support Agent, and Read-Only Viewer permissions enforced at both route middleware and Blade template level.
- Built RESTful API using Laravel API resources with Fractal transformers providing sparse fieldsets, eager-loading via ?include= parameter, and standardised JSON:API envelope consumed by Paddle's React frontend.
- Improved financial report generation performance by 35% using MySQL materialised reporting table populated by nightly Artisan command computing MRR, ARR, churn rate, and revenue recognition per billing period.

PROJECT 2: ShopKart – WooCommerce Alternative E-Commerce | Domain: E-Commerce

Client: Shopmatic (Singapore-based SMB e-commerce SaaS) | **Domain:** E-Commerce / Multi-Channel Retail

Tech Stack: PHP, CodeIgniter, MySQL, jQuery, AJAX, Razorpay, PayPal, Bootstrap, Git, cPanel

Description: Full-featured multi-vendor e-commerce platform enabling 25,000+ Southeast Asian merchants to manage product catalogues, process orders, handle multi-currency payments, and track shipments across sales channels.

Roles & Responsibilities:

- Designed CodeIgniter MVC architecture with modular controllers for Catalogue, Cart, Order, Payment, and Shipping domains enabling parallel development by 4 developers without merge conflicts on shared codebase.
- Built dynamic product catalogue with hierarchical category management, multi-attribute variant system (colour × size × material), bulk import via CSV with validation, and image upload to AWS S3 via pre-signed URL.
- Implemented AJAX-driven shopping cart using JavaScript session + database dual persistence ensuring cart state survives browser refresh, device switch, and guest-to-authenticated user conversion without loss.
- Integrated Razorpay and PayPal payment gateways with server-side order verification, webhook receiver validating payment.captured events, and idempotent order status updates preventing duplicate fulfilment.
- Developed admin dashboard with jQuery DataTables for order management, AJAX-based status updates, and CodeIgniter-generated Excel export of sales reports using PHPEXcel with conditional formatting.
- Built product search using MySQL FULLTEXT index on (product_name, description, sku) with MATCH AGAINST BOOLEAN mode search reducing search response time from 1.8 seconds to 95 milliseconds.
- Implemented XSS prevention using htmlspecialchars() output escaping, CSRF token validation on all POST forms, prepared statement binding on all MySQL queries, and rate limiting on payment endpoints.
- Designed order management workflow: Pending → Processing → Shipped → Delivered → Returned with email/SMS notification via Twilio at each status transition and automated refund initiation on return approval.
- Optimised MySQL query performance using EXPLAIN ANALYSE on 15 slow queries adding composite indexes and rewriting subquery-heavy reports as JOIN-based queries reducing report generation from 9 seconds to 1.2 seconds.
- Deployed on cPanel Linux with OPcache enabled, MySQL slow query log monitoring, and automated Git-based deployment script triggering composer install and artisan cache:clear on every release push.

PROJECT 3: CareRecords – DrChrono-Style Healthcare SaaS | Domain: Healthcare

Client: Huma (UK-based digital health SaaS) | **Domain:** Healthcare / Patient Management

Tech Stack: PHP, Laravel, MySQL, REST API, Redis, Twilio, AWS S3, Bootstrap, Git, Linux

Description: HIPAA-aligned healthcare SaaS platform managing electronic health records, appointment scheduling, prescription management, billing workflows, and lab result delivery for 200+ clinic and hospital clients.

Roles & Responsibilities:

- Implemented Laravel Sanctum ability-scoped API tokens with distinct abilities for doctor, nurse, receptionist, and admin roles enabling fine-grained PHI access control per token issued to clinic staff.
- Built appointment scheduling engine with Laravel queue-based slot reservation (optimistic locking on appointment_slots table using MySQL SELECT FOR UPDATE) preventing double-booking under concurrent clinic traffic.
- Integrated Twilio Programmable SMS for appointment reminder dispatch: Laravel scheduled job queries upcoming appointments 24 hours ahead and dispatches reminder job to Horizon queue with retry on delivery failure.
- Designed Laravel Eloquent models with field-level AES-256 encryption via laravel-encryption package for PHI columns (patient_ssn, diagnosis_code, prescription_details) using AWS KMS-rotated encryption keys.
- Built prescription management module with drug interaction checking via external API, dosage history tracking, digital signature capture stored as Base64 PNG to AWS S3, and PDF generation via Laravel DomPDF.
- Developed billing module with insurance claim submission via HL7 FHIR REST API, ERA payment posting, denial management workflow, and patient statement generation with outstanding balance calculation.
- Implemented Laravel Observer pattern for EHR audit trail: PatientObserver fires on every create/update/delete attaching user_id, IP address, changed_fields JSON, and timestamp to immutable audit_log table.
- Optimised complex patient history SQL queries using covering indexes on (patient_id, visit_date, doctor_id) and MySQL query cache reducing patient record load from 3.4 seconds to 180 milliseconds under load.
- Built RESTful API with Laravel API resources for third-party EHR integrations providing FHIR-compliant Patient, Encounter, and MedicationRequest resource endpoints consumed by partner hospital management systems.
- Deployed on Linux VPS with Supervisor managing Laravel queue workers, Redis caching of frequently accessed patient demographics, and automated daily backup of MySQL database to encrypted AWS S3 bucket.

PROJECT 4: EduPortal – Learnosity-Style Education SaaS | Domain: Education

Client: Learnosity (Ireland-based EdTech SaaS) | **Domain:** EdTech / Learning Management

Tech Stack: PHP, Laravel, MySQL, Redis, REST API, Bootstrap, jQuery, AJAX, AWS S3, Git

Description: Multi-tenant education SaaS platform supporting 300+ institutions to manage student enrolment, course delivery, assessment workflows, progress tracking, certificate generation, and learning analytics.

Roles & Responsibilities:

- Architected multi-tenant education platform using Laravel with database-per-tenant isolation strategy providing 300+ institutions fully separate MySQL databases provisioned on new client onboarding via Artisan command.
- Built adaptive assessment engine: question difficulty levels stored per question record, student performance history queried via Eloquent, and next question selected via PHP algorithm adjusting difficulty based on rolling score.

- Implemented Laravel Scout full-text course and resource search with Meilisearch delivering sub-60ms typeahead results across 200,000+ course materials, questions, and student records per tenant.
- Developed certificate generation pipeline: Laravel job triggered on course completion computes final grade, generates personalised PDF certificate via TCPDF with QR code linking to verification API, stores on S3.
- Built progress analytics dashboard using MySQL window functions (LAG, LEAD, RANK) and Laravel Eloquent raw expressions generating student engagement reports, at-risk student alerts, and cohort performance summaries.
- Integrated video content delivery using AWS S3 signed URLs with 1-hour expiry preventing unauthorised direct access to course video files and enabling time-limited streaming via HLS without media server.
- Implemented online exam proctoring features: random question ordering per attempt, time-limited sessions enforced server-side via Redis TTL, tab-switch detection via JavaScript blur event logging to violation_log table.
- Developed RESTful API with Laravel Passport OAuth2 for mobile app integration providing course content, assessment submission, progress sync, and push notification trigger endpoints for iOS/Android student apps.
- Optimised database performance adding composite indexes on (tenant_id, student_id, course_id, completion_status) and MySQL query cache reducing student dashboard load from 4.1 seconds to 310 milliseconds.
- Built automated billing integration connecting EduPortal to Razorpay subscription API: institution onboarding triggers plan creation, monthly invoice auto-generated via webhook, and access revoked on payment failure.

PROJECT 5: ERPCore – Zoho-Alternative Business Management | Domain: ERP / Manufacturing

Client: Katana MRP (Estonia-based manufacturing ERP SaaS) | **Domain:** ERP / Manufacturing & Inventory

Tech Stack: PHP, CodeIgniter, MySQL, AJAX, jQuery, REST API, Bootstrap, Git, Linux/cPanel

Description: Comprehensive ERP platform for manufacturing SMBs managing inventory, bill of materials (BOM), production orders, purchase management, sales workflows, and financial reporting for 1,500+ business clients.

Roles & Responsibilities:

- Designed MySQL inventory schema with stored procedures for FIFO, LIFO, and weighted average cost valuation methods computing accurate cost of goods sold and inventory valuation across multi-warehouse operations.
- Built bill of materials (BOM) management module with nested CodeIgniter recursive component tree supporting multi-level assemblies, phantom BOM items, and automated raw material requirement computation per production order.
- Developed real-time production order tracking dashboard using AJAX long-polling updating machine utilisation, work-in-progress (WIP) status, and completion percentage without full page reload every 30 seconds.
- Implemented purchase order workflow with three-way matching: purchase order → goods receipt → supplier invoice matched by CodeIgniter validation service blocking payment if quantities or amounts deviate beyond threshold.
- Built AJAX-driven sales order management with available-to-promise (ATP) check querying MySQL inventory in real-time during order entry preventing customer promises on out-of-stock finished goods.
- Developed financial reporting module with MySQL complex JOIN queries generating P&L, balance sheet, cash flow statement, and aged receivables report with PHPEXCEL export and PDF generation via mPDF.

- Designed role-based access with CodeIgniter ACL library granting module-level read, write, approve, and export permissions per user role (Warehouse Manager, Sales Rep, Accountant, Director, Admin) enforced at controller level.
- Optimised 18 slow MySQL queries using EXPLAIN ANALYSE: added covering indexes, rewrote correlated subqueries as LEFT JOIN, and replaced DISTINCT on large sets with GROUP BY reducing ERP dashboard load by 65%.
- Integrated third-party accounting APIs (Tally, QuickBooks) via REST webhooks synchronising journal entries, ledger postings, and financial period close data bidirectionally without manual data entry duplication.
- Deployed on cPanel Linux with PHP OPcache, MySQL query cache tuning, and Supervisor-managed background worker processing nightly inventory valuation, report pre-computation, and automated reorder point alerts.